Lecture Notes in Electrical Engineering 966

N. Subhashini
Morris. A. G. Ezra
Shien-Kuei Liaw *Editors*

Futuristic Communication and Network Technologies

Select Proceedings of VICFCNT 2021, Volume 1

Springer

N. Subhashini · Morris. A. G. Ezra ·
Shien-Kuei Liaw
Editors

Futuristic Communication and Network Technologies

Select Proceedings of VICFCNT 2021, Volume 1

Springer

Modified Hill Cipher with Invertible Key Matrix Using Radix 64 Conversion

A. Ashok Kumar, S. Kiran, and D. Sandeep Reddy

Abstract Cryptography is one of the most important technological areas which will provide security to the data by using various cryptographic algorithms. The Hill cipher (HC) is a known symmetric encryption algorithm using linear matrix transformation. In spite of the simplicity of Hill cipher algorithm, there are several advantages such as masquerading letter frequencies of the plaintext and high throughput. The traditional Hill cipher founds a serious setback due to the vulnerability against known plaintext–ciphertext attacks. So, to enhance security, a new variant of the Hill cipher method is proposed. The proposed algorithm develops in three stages, firstly, convert the plaintext into 8-bit binary form; in the second stage, we remove the last two bits from MSB positions as they are 0 in the MSB and 1 in the MSB-1 position, respectively, and this reduces the size of the data. Now, apply Radix 64 conversion to each row; in the last stage, we perform the traditional Hill cipher algorithm (modulo 64), where we use the invertible matrix as key, and this produces the ciphertext. The proposed technique reduces the size of ciphertext data after encryption. The security analysis of the modified algorithm using avalanche effect shows three-fold increase than comparing to tradition Hill cipher algorithm. Further, the data size dependence on the performance of the modified Hill cipher algorithm is also addressed and found that the encryption time is increased with increase of data size.

Keywords Information security · Radix 64 conversion · Invertible matrix · Hill cipher · Cryptography · The CIA triad

A. A. Kumar
Physics, YSR Engineering College of YVU, Proddatur, AP, India

S. Kiran (✉)
Department of CSE, YSR Engineering College of YVU, Proddatur, AP, India
e-mail: rkirans125@gmail.com

D. S. Reddy
ISSE, NIT Jamshedpur, Jharkhand 831014, India

N. Subhashini et al. (eds.), *Futuristic Communication and Network Technologies*, Lecture Notes in Electrical Engineering 966,
https://doi.org/10.1007/978-981-19-8338-2_14

1 Introduction

There is rapid growth in the economy of the world due to digitization in every field from e-papers to Bitcoin transactions with the help of the Internet [1]. Because of digital communication, there is a rapid increase in usage of the Internet from the past few decades. Data security is the primary objective in the processing of digital data [2, 3]. To provide security for the data, the data are to be transmitted through a secure medium. This can be achieved by using suitable cryptographic algorithms [1–4]. Cryptography is a branch of science and technology which uses mathematical algorithms to encrypt and decrypt information [5]. The main aim of using different algorithms in cryptography is to enable us to provide security to the sensitive information, which preferably transmits in insecure channels like Internet. The data cannot be protected from being accessed by attackers instead the data can be made unreadable during transmission, and only the intended recipient can reconvert the unreadable to a readable form [4]. The process of converting the plaintext into an unreadable format (ciphertext) using any cryptographic algorithm is known as encryption. The ciphertext which can be decoded back into the original message is known as decryption [5]. Along with the cryptographic algorithm, a key may be used to encode or decode the plaintext; if the same key is used for both encryption and decryption, then that algorithm is known as the symmetric-key algorithm. If not, then it is known as the asymmetric-key algorithm [4].

The dependence of cryptographic protection algorithms of a system against attacks and unauthorized user penetration depends on many factors. Some of the important factors includes the strengthening of the keys and effective management of its associated protocols. It is also essential to develop algorithms such as secure key generation, storage, distribution, use, and destruction to enhance the protection of the keys used in the encryption process. Several effective symmetric key algorithms have been developed over a past decade which include Blowfish, DES, RC5, AES, RC6, etc. One method that is often used is a Hill cipher algorithm which refers to symmetric cryptosystems which largely relies on matrix operations [6]. Different approaches have been developed to produce different variants of Hill cipher algorithms to strengthen its security mechanisms. Several other specific matrices, such as Vandermonde matrix [7] and orthogonal matrix [8], have been proposed as key matrices. The applications of Hill cipher are not only limited to cryptography but can also be extended to encrypt biometric signals [9].

The use of matrices in cryptography is started with the invention of the Hill cipher algorithm, which is a polygraphic substitution cipher based on linear algebra [3, 10]. Each letter is represented by a number modulo 26. Often, the simple scheme $A = 0$, $B = 1$, …, $Z = 25$ is used, but this is not an essential feature of the cipher. The key in this algorithm is a $n*n$ non-singular matrix [i.e., det(matrix)! = 0], then only the inverse for the matrix exists. Before that, cryptographic algorithm is using the same key for encryption and decryption, but in the case of the Hill cipher algorithm, same key is shared with the recipient, but for decryption, the inverse matrix of the key is used. The data that are transferred (irrespective of medium) travel in the form of bits, not in the same way as they are sent. As a reason for this, we will be mostly concentrating on the bit operations in this paper. The matrix used for the encryption method is called the encoding matrix, and the matrix used for decryption is called the decoding matrix. An extension to Hill cipher is provided in this paper which helps in reducing the data to be shared when compared with the data given. This is possible with the help of Radix 64 conversion, where two bits are removed from every 8 bits to reduce the data and the two bits are 0 in the MSB position and 1 in the MSB-1 position; this is observed in all the cases; if not, a possible alternative is chosen to correct this technique.

2 Literature Survey

The main limitation of the traditional Hill cipher algorithm is that it is prone to a known-plaintext attack. If the attacker has distinct pairs of plaintext and ciphertext, then the key value can be retrieved by solving the plaintext and ciphertext. Amit Kumar Mandle et al. [1] and Maheswari et al. [5] and other authors introduced various variants of the Hill cipher algorithm for encryption and decryption of messages. But, they have a limitation as it works only for the alphabets and space but not for other characters. For encryption and decryption, no single algorithm is sufficient to provide security. As researchers are working on it to provide greater security with the inclusion of different methods by modifying the Hill cipher algorithm. Aaref et al. [11] proposed a new hybrid secured approach of ciphering by using substitution followed by transposition cipher methods. They reported that the proposed method (Rail Hill) is highly secured and difficult to break and acts as a bridge between classical and modern ciphers. In this hybrid approach, they designed Hill cipher using substitution cipher algorithm and transposition cipher algorithm is used in rail fence. After evaluating the execution time, they found that the proposed algorithm is more secured and difficult to break than tradition substitution and transposition algorithms. Bakr et al. [12] proposed a modification on the Elliptic Curve Cryptography. This modification shows lesser computations' time and further improves the performances of security.

Lot of groups worked on modification of Hill cipher using matrix operations such as non-invertible key matrix [13] where the users do not have difficulty to identify inversion of a key matrix when the key matrix is invertible for decryption process, used rectangular matrix key is created using Playfair Cipher algorithm [14] to improve the security of the Hill cipher algorithm, genetic algorithms [15] have also been used to evaluate the correct key both for encryption and decryption processes of the Hill cipher algorithm, modified S-Box [16] is also utilized in the encryption process of Hill cipher algorithm to enhance the security, similarly P-box and M-Box techniques [17] are used to enhance the security issues of Hill cipher algorithm like avalanche effect and also demonstrated the data dependence of the algorithm, a block cipher is produced by introducing interweaving and iteration functions [18] because of which the plaintext has undergone several transformations before it has become the ciphertext which significantly enhances the security, efficient methods are generated to generate self-invertible matrix [19] for Hill cipher algorithm where the methods are less computational complexity, a known plaintext attack of Hill cipher is suppressed by sharing a prime circulant key as secret key and non-singular matrix 'G' as public key [20], of which the determinant of G_C gives 0 which generates infinite solution, and hence, security of this algorithm is enhanced.

3 Hill Cipher Algorithm

Hill cipher algorithm is a polyalphabetic substitution cipher based on linear algebra [21]. It was the first polygraphic algorithm that converts three or more characters of the plaintext at a time [3, 10]. In the traditional Hill cipher algorithm, each character is assigned with an integral value as **A = 1, B = 2……Z = 26, Space = 0** [5]. The existing algorithm is explained in the below-defined successive steps. Firstly, the plaintext is divided into column matrices of size **m*1,** and **m** is the size of the square invertible matrix **(m*m)**. The second step is to convert the plaintext into its respective integer value. This method is applicable for only the specified 27 characters and generates the ciphertext in the same range, where the plaintext is defined. To achieve the resulted values in the same range, the existing algorithm uses MODULAR DIVISION (mod 27); here, 27 defines the total number of characters in the specified range. Then, construct an m*m square invertible matrix. To determine a square matrix as an invertible matrix, it must satisfy the condition '**det(matrix) = 0**'. The key value must have an invertible matrix such that the product of the key and its inverse is an identity matrix ($\boldsymbol{K} * \boldsymbol{K}^{-1} = \mathbf{I}$). Now, perform matrix multiplication with key value and with column matrix one at a time and perform **mod 27** for the intermediate result (result generated after matrix multiplication). Convert the resulted values into the text form, and thus, generated unreadable text is the ciphertext.

Encryption can be mathematically defined as

$$C = K * P(\mathrm{mod}27).$$

Here

- C is the ciphertext formed,
- K is the key matrix,
- P is the plaintext matrix.

Decryption can be mathematically defined as

$$P = I * C(\mathrm{mod}27).$$

Here

- P is the plaintext formed,

Table 1 Conversion table of character to a respective integer value

A	1	**J**	10	**S**	19
B	2	**K**	11	**T**	20
C	3	**L**	12	**U**	21
D	4	**M**	13	**V**	22
E	5	**N**	14	**W**	23
F	6	**O**	15	**X**	24
G	7	**P**	16	**Y**	25
H	8	**Q**	17	**Z**	26
I	9	**R**	18	**SPACE**	0

I is the inverse of the key matrix

$$\left(I = \boldsymbol{K}^{-1} = \frac{\boldsymbol{Adj}(\boldsymbol{K})}{\boldsymbol{det}(\boldsymbol{K})}\right),$$

C is the ciphertext.

Explanation
Let us consider the plaintext as: '**HILL CIPHER ALGORITHM**'.

- Convert the plaintext into integer value with the reference of Table 1.

H	I	L	L		C	I	P	H	E	R		A	L	G	O	R	I	T	H	M
8	9	12	12	0	3	9	16	8	5	18	0	1	12	8	15	18	9	20	8	13

- Convert the plaintext into the columnar matrix of size 3*1.

$$P1 = \begin{bmatrix} 8 \\ 9 \\ 12 \end{bmatrix}, P2 = \begin{bmatrix} 12 \\ 0 \\ 3 \end{bmatrix}, P3 = \begin{bmatrix} 9 \\ 16 \\ 8 \end{bmatrix}, P4 = \begin{bmatrix} 5 \\ 18 \\ 0 \end{bmatrix},$$

$$P5 = \begin{bmatrix} 1 \\ 12 \\ 7 \end{bmatrix}, P6 = \begin{bmatrix} 15 \\ 18 \\ 9 \end{bmatrix}, P7 = \begin{bmatrix} 20 \\ 8 \\ 13 \end{bmatrix}$$

- Construct a key matrix $K = \begin{bmatrix} 4 & -2 & 3 \\ 8 & -3 & 5 \\ 7 & -2 & 4 \end{bmatrix}$.

- Now, multiply each columnar matrix with the key value, and the resulting value is the intermediate result. Apply mod 27 to the intermediate result. Convert the numerical values to text format with the reference of Table 1.

$$C1 = K * P1 = \begin{bmatrix} 4 & -2 & 3 \\ 8 & -3 & 5 \\ 7 & -2 & 4 \end{bmatrix} * \begin{bmatrix} 8 \\ 9 \\ 12 \end{bmatrix} = \begin{bmatrix} 50 \\ 97 \\ 86 \end{bmatrix} (\text{mod} 27) = \begin{bmatrix} 23 \\ 16 \\ 5 \end{bmatrix} = \begin{bmatrix} W \\ P \\ E \end{bmatrix},$$

$$C2 = \begin{bmatrix} 3 \\ 3 \\ 15 \end{bmatrix} = \begin{bmatrix} C \\ C \\ I \end{bmatrix}, C3 = \begin{bmatrix} 1 \\ 10 \\ 9 \end{bmatrix} = \begin{bmatrix} A \\ J \\ I \end{bmatrix}, C4 = \begin{bmatrix} 11 \\ 13 \\ 26 \end{bmatrix} = \begin{bmatrix} K \\ M \\ Z \end{bmatrix}$$

$$C5 = \begin{bmatrix} 1 \\ 7 \\ 11 \end{bmatrix} = \begin{bmatrix} A \\ G \\ K \end{bmatrix}, C6 = \begin{bmatrix} 24 \\ 3 \\ 24 \end{bmatrix} = \begin{bmatrix} X \\ C \\ X \end{bmatrix}, C7 = \begin{bmatrix} 22 \\ 12 \\ 14 \end{bmatrix} = \begin{bmatrix} V \\ L \\ N \end{bmatrix}.$$

- The ciphertext formed is ‘**WPECCIAJIKMZAGKXCXVLN**’.
- The reverse of the encryption technique is the decryption technique.

This traditional Hill cipher alone is not secure in its respective domain. So, to attain greater complexity of the ciphertext, different techniques are combined with the existing technique. In the proposed technique, a two stage hybrid approach was utilised includes removing common consecutive columns in each block and applying Radix 64 conversion.

4 Proposed Methodology

The traditional Hill cipher has several advantages in its respective domain. It is a symmetric key matrix but does not use the same key for the encryption and decryption processes [5]. It generates an invertible matrix for the decryption process with the use of a key matrix. The main reason to develop this proposed methodology is that it reduces the data that are to be transferred when compared with the data that are given. The proposed methodology is described in four stages. Firstly, the plaintext characters are substituted. In the second stage, convert the plaintext to 8-bit binary form. In the third stage, remove MSB and MSB-1 bits as they are 0 and 1. Now, the 8-bit value is reduced to 6-bit value and convert them to decimal form. In the fourth stage, perform the traditional Hill cipher algorithm along with modulo 64 to generate the ciphertext in the range of the Radix 64 conversion table. The reverse procedure of encryption is decryption.

4.1 Radix 64 Conversion

Radix 64 conversion is the method of representing the characters in a 6-bit binary value concerning the Radix 64 conversion table. The Radix 64 conversion table consists of 64 characters. It consists of 26 uppercase characters (A–Z), 26 lowercase characters (a–z), ten digits (0–9), and two special characters (‘ + ’ and ‘/’). First, each character of the plaintext is converted into its respective ASCII value. After the removal of two bits from each 8-bit binary value, there will be 6 bits remaining, and the 6 bits left are in the Radix 64 form. The process of converting character into its respective 6-bit binary value is called Radix 64 encoding, and the reverse process of converting the 6-bit binary value to its character is called Radix 64 decoding.

Algorithm of Encryption

1. Rewrite the plain text after necessary replacement of characters i.e., SPACE is replaced by ‘_’, ‘.’ is replaced by ‘|’ and ‘,’ is replaced by ‘\’.
2. Convert the plain text into ASCII values and then to 8-bit binary form.
3. Remove the last two bits from MSB positions as they are ‘0’ in the MSB and ‘1’ in the MSB-1 position respectively from every 8-bit binary value.
4. The 6 bits left after the removal of 2-bits are converted into Radix 64 value.
5. Group these values into columnar matrices of size 3*1 or 4*1 or more.
6. Apply traditional Hill Cipher and apply modulo 64 to the result and convert the resulted values into respective Radix 64 conversion.
7. The resulted text is the Ciphertext.

Algorithm of Decryption

1. Convert the ciphertext into its respective Radix 64 value & group these values into columnar matrices of size 3*1 or 4*1 or more.
2. Apply traditional Hill Cipher and apply modulo 64 to the result.
3. Convert the above-resulted values into Radix 64 Encoding (6-bit).
4. Add two bits to each 6-bit binary value i.e., ‘0’ at MSB position and ‘1’ at the MSB-1 position.
5. Convert the 8-bit binary value into its ASCII value to character.
6. Now replace ‘_’ with SPACE, ‘|’ With ‘.’ And ‘\’ with ‘,’.
7. The resulted text is the plain text.

Table 2 Various sizes of data after encryption and decryption

S. No.	Data size(in bits)	After encryption (in bits)	After decryption (in bits)
1	64	48	64
2	128	96	128
3	256	192	256
4	512	384	512
5	1 Kb	768	1024
6	2 Kb	1536	2048
7	4 Kb	3072	4096
8	5 Kb	3840	5120
9	10 Kb	7680	10,240

5 Performance and Security Analysis

5.1 Analysis of Data Size After Encryption and Decryption Processes

From the analysis, it is observed that the data size is reduced after doing encryption process. The data is retrieved back to the same original size after the decryption process. Table 2 represents how the size of the data is varied with respect to the encryption and decryption mechanisms.

5.2 Avalanche Effect

The security of the proposed algorithm was analyzed using avalanche effect. It was found that the change of 1 bit in the plaintext results in change of three characters irrespective of size. It was also analyzed that the change of 1 bit in the key results in 33.3% change of characters in the ciphertext.

5.3 Encryption Time Dependence on File Size

The encryption and decryption times were evaluated for the proposed algorithm for different sizes of the file. It is evidenced that the encryption time increases when file size increases but at a lower rate than compared to the file size. The statistical analysis is mentioned in Table 3.

From the above analysis, it is observed that security of the modified Hill cipher using Radix 64 improves more than three folds comparing to original Hill cipher

Table 3 Variation of encryption time with file size

Size of file (in bytes)	Encryption execution time (in ms)	File size (in KB)	Encryption execution time (in ms)
64	424 ms	1 KB	799 ms
128	442 ms	2 KB	915 ms
256	476 ms	3 KB	1120 ms
512	557 ms	4 KB	1441 ms
1024	799 ms	5 KB	1542 ms

algorithm [16]. The proposed variant of Hill cipher algorithm may be used than tradition Hill cipher in view of the advantages mentioned in the performance analysis.

6 Conclusion

A new variant of the Hill cipher algorithm is proposed using Radix 64 conversion. The proposed algorithm encrypts and decrypts any type of message not only alphabets and characters. The ciphertext generated using this algorithm is also in the same range as the plaintext. The size of the ciphertext formed using this proposed methodology is smaller when compared with the size of the plaintext. Because of the decrease in the size of the ciphertext after encryption, the complexity of breaking the ciphertext is difficult. The proposed methodology generates 48 bits of ciphertext for every 64 bits of plaintext. The security analysis and data size dependence on the performance of the modified Hill cipher algorithm are also addressed. Avalanche effect of the proposed algorithm shows 33.3% change of characters in the ciphertext for every 1-bit change in the key. Increase in the change of encryption time is observed with increase of file size but at a lower rate.

References

1. Mandle AK, Namdeo V (2019) Encryption and decryption of a message involving byte rotation technique and invertible matrix. Int J Eng Adv Technol (IJEAT), ISSN: 2249–8958, 9(2):1160–1163
2. Reddy SK, Kishore RS (2014) A new digital encryption scheme matrix rotations and bytes conversion encryption algorithm. Int J Eng Res Technol (IJERT) ISSN: 2278–0181 3(7):487–494
3. Kalaichelvi V, Manimozhi K, Meenakshi P, Rajkumar B, Vimala Devi P (2017) A New variant of hill cipher algorithm for data security. Int J Pure Appl Math. ISSN: 1311–8080, 117(15):581–588
4. Kumar BR, Murti PRK (2011) Data encryption and decryption process using bit shifting and stuffing (BSS) methodology. Int J Comput Sci Eng (IJCSE) ISSN: 0975–3397, 3(7):2818–2827

5. Maheswari D, Kaushika A, Jenifer A (2018) A study on data encryption and decryption using hill cipher algorithm. Int J Tech Innov Modern Eng Sci (IJTIMES). e-ISSN: 2455–2585, 4(3):21–26
6. Liew KJ, Nguyen VT (2020) Hill cipher key generation using skew-symmetric matrix. In: Proceedings of the 7th international cryptology and information security conference 2020 (CRYPTOLOGY2020)
7. Sharma P, Rehan M (2014) Modified hill cipher using vandermonde matrix and finite field. Int J Technol 4(1):252–256
8. Khan FH, Shams R, Qazi F, Agha D (2015) Hill cipher key generation algrithm by using orthogonal matrix. Proc Int J Innov Sci Modern Eng 3(3):5–7
9. Kaur H, Khanna P (2017) Non-invertible biometric encryption to generate cancelable biometric templates. Proc World Congress Eng Comput Sci 1:1–4
10. Siahaan MDL, Siahaan APU (2018) Application of hill cipher algorithm in securing text messages. Int J Innov Res Multidisciplin Field. 4(10):55–59, ISSN: 2455–0620
11. Aaref AM, Ablhd AZ (2017) A new cryptography method based on hill and rail fence algorithms. Diyala J Eng Sci 10(1):39–47
12. Bakr MA, Mokhtar A, Takieldeen A (2018) Elliptic curve cryptography modified hill cipher dependent on circulant matrix. Int J Indus Electron Electric Eng 6(7):24–29
13. Saxena MA, Mr. Lohiya H, Mr. Patidar K (2017) A novel technique of hill cipher for evaluation of non-invertible key matrix 06(1):856–862
14. Alawiyah T, Hikmah AB, Wiguna W, Kusmira M, Sutisna H, Simpony BK (2020) Generation of rectangular matrix key for hill cipher algorithm using playfair cipher. J Phys: Conf Series 1641 012094:1–5
15. Siahaan APU (2016) Genetic algorithm in hill cipher encryption. Am Int J Res Sci Technol Eng Math 15(1):84–89
16. Paragas JR, Sison AM, Medina RP (2019) A new variant of hill cipher algorithm using modified S-box. Int J Sci Technol Res 8(10):615–619
17. Khar S, Bhargawa N, Shukla R, Shukla M (2012) Implementation of enhanced modified hill cipher by P-box and -M-box technique. Int J Inf Technol Knowl Manag 5(1):53–58
18. Sastry VU, Shankar NR, Bhavani SD (2010) A modified hill cipher involving interweaving and iteration. Int J Netw Secur 11(1):11–16
19. Acharya B, Rath GS, Patra SK, Panigrahy SK (2007) Novel methods of generating self-invertible matrix for hill cipher algorithm. Int J Secur 1(1):14–21
20. Reddy KA, Vishnuvardhan B, Madhuviswanatham, Krishna AVN (2012) A modified hill cipher based on circulant matrices. Procedia Technol 4:114–118
21. Abidin AFA, Chuan OY, Ariffin MRK (2011) A novel enhancement technique of the hill cipher for effective cryptographic purposes. J Comput Sci 7(5):785–789